Intra Puff Pressure Analyzer

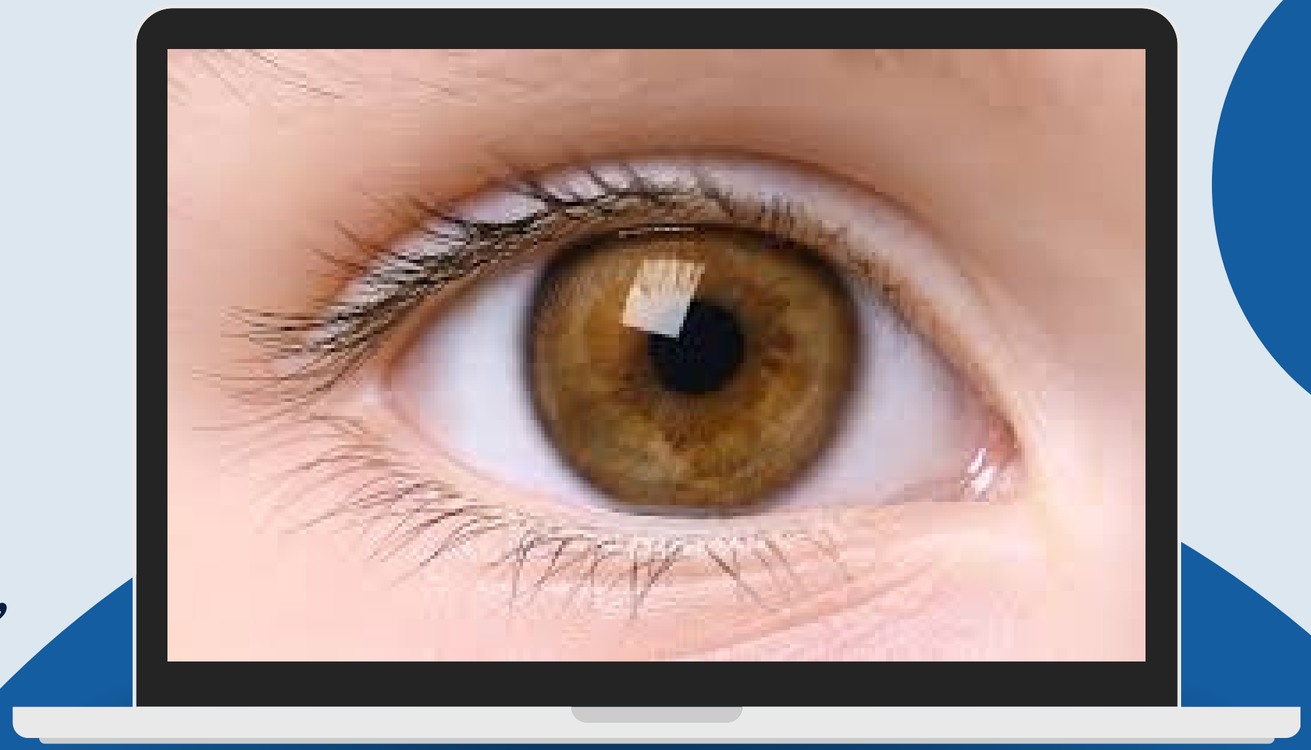
TEAM MEMBERS:

SARAVANAN R K

VASANTH K

ARAVINDSELVAN C

COLLEGE NAME : SRI SAIRAM ENGINEERING COLLEGE, CHENANI



Overview

- Who Are We? 04
- Problem/Oppurtunity 05
- Interview & Survey 06
- Market size estimation 08
- Customer Persona 10
- Value Proposition Canvas 11
- Solution 12
- Lean Canvas 13
- Competion Analysis 14
- MVP Description 17
- MVP Validation 22
- Branding, Positioning and Channels 24
- Go-To- Market Strategy 25
- Sales Funnel 27
- Customer Acquisitionn Plan 28
- Start-up Costs 30
- Forecast Profit & Loss 31
- Fundin Plan & Projections 32
- Breakeven Analysis 33
- Unit Economics 34
- Funding Plan 35
- Brand Framework 36
- Team Composition 37



About Us

Our venture aims to address the challenges faced by patients by developing a portable, non-invasive device that utilizes air puff technology to measure IOP. This solution will offer a more patient-friendly approach, reducing discomfort while enabling frequent monitoring outside of clinical environments. By making glaucoma detection and IOP measurement more accessible, our device has the potential to revolutionize early diagnosis and long-term management of the disease, ultimately reducing the global burden of blindness caused by glaucoma.

We are AirTon



Why

Glaucoma is a leading cause of blindness, and its early detection depends on accurate measurement of intraocular pressure (IOP). Existing methods are invasive, uncomfortable, and limited to clinical environments, making routine monitoring difficult and inaccessible to many.



What

It is a portable, non-invasive device that uses air puff technology to measure IOP. It provides a comfortable, patient-friendly method for early glaucoma detection and regular monitoring outside clinical settings.



How

The portable device measures intraocular pressure (IOP) using a gentle air puff to deform the cornea, with pressure sensors analyzing the deformation. IoT integration enables remote monitoring, and interference pattern analysis ensures accuracy. It offers convenient self-monitoring at home for early detection of eye problems. We generate revenue by selling the device to hospitals, eyecare clinics and we also provide on the basis of subscription

Problem / Opportunity

Context
When does the problem occur

- Glaucoma is a leading cause of irreversible blindness worldwide, characterized by elevated intraocular pressure (IOP) that damages the optic nerve.
- Current methods for measuring IOP are invasive, uncomfortable, and predominantly limited to clinical settings.
- This restricts access to timely and frequent monitoring, particularly for people in remote or underserved areas.

Problem
What is the root cause of the problem

Traditional IOP measurement techniques, such as Goldmann applanation tonometry, require direct contact with the eye. This process is uncomfortable, can cause infection, and must be performed by trained professionals, restricting the frequency of testing. The lack of portable, non-invasive solutions further exacerbates accessibility challenges, hindering early detection and monitoring of glaucoma.

Alternatives
What do customers do now to fix the problem

- Contact tonometry methods
- Handheld tonometers
- Home-use devices with limited non-invasive capabilities

Customers
Who has the problem most often

- **Primary:** Glaucoma patients, especially those requiring regular monitoring of IOP.
- **Secondary:** Eye care professionals, clinics, hospitals, and telemedicine platforms looking for more accessible, patient-friendly solutions.
- **Tertiary:** Remote or underserved populations with limited access to clinical facilities and specialized eye care.

Emotional Impact

Alleviate fear, improve patient compliance, and provide a sense of empowerment and reassurance by making regular monitoring accessible and stress-free.

Quantifiable Impact

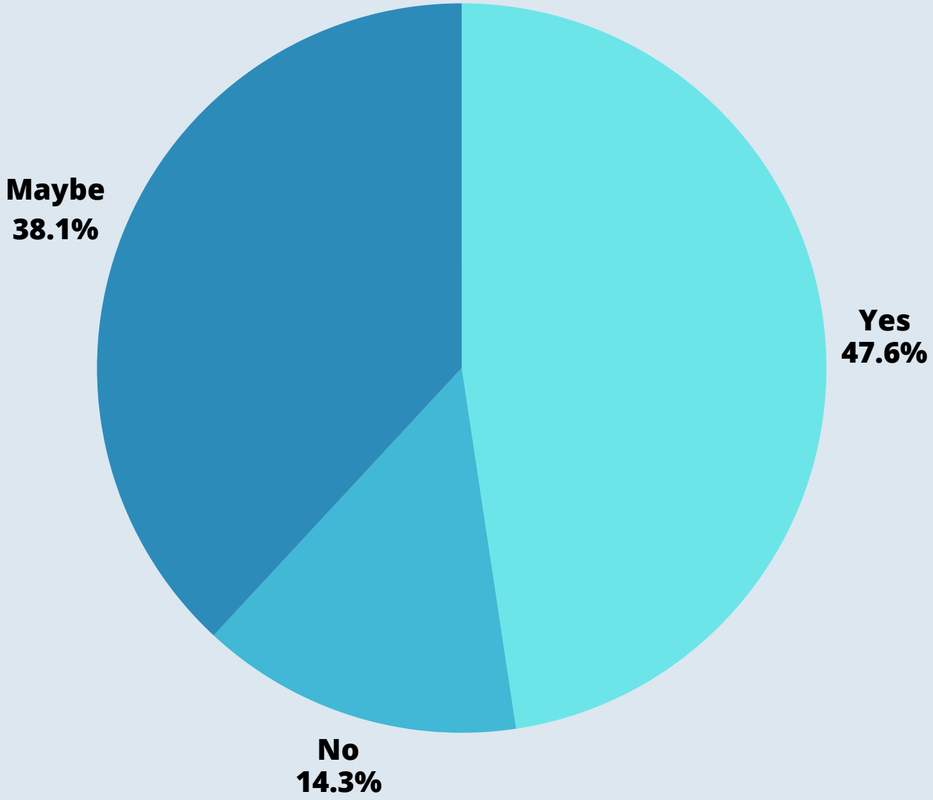
- Increased Screening Rates
- Cost Reduction
- Enhanced Quality of Life
- Improved Patient Compliance

Alternative Shortcomings
What are the disadvantages of the alternatives

- Discomfort
- Accuracy and accessibility
- Costly
- Require specialized training,
- Unsuitable for widespread home or field use.

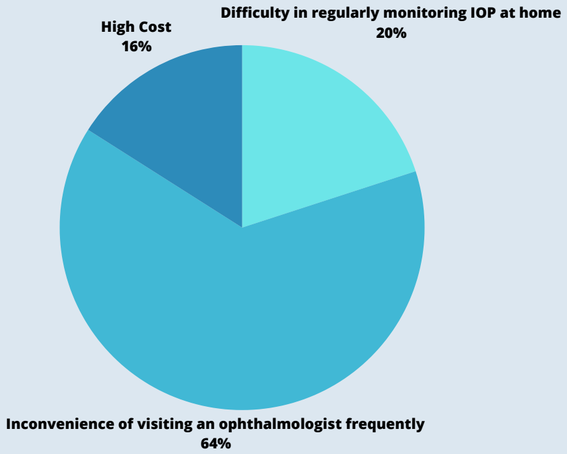
Interview & Survey Results

- How many customers did you interview? - 21
- What was the interview mode? - Google Forms
- How many of them agree this is a problem and wants a solution? - 10
- How many of them said they may need a new solution? - 8
- How many of them said they don't need a new solution? - 3

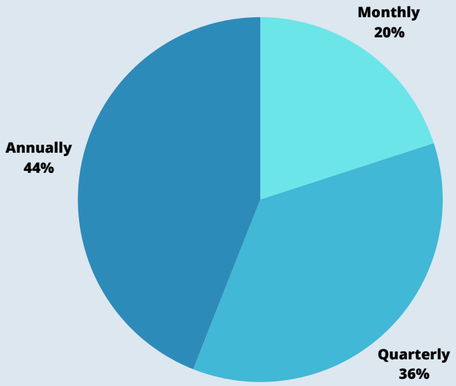


- Google forms link: https://docs.google.com/forms/d/e/1FAIpQLSdIN_PjkWh2PDeRRQOI5K0dyzcQX_Zm9_xsOYMeHcozz-huFg/viewform?usp=sf_link

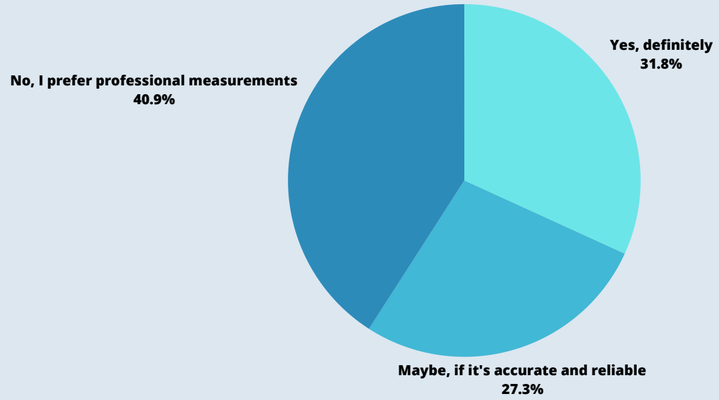
1. What is your biggest challenge in managing glaucoma?



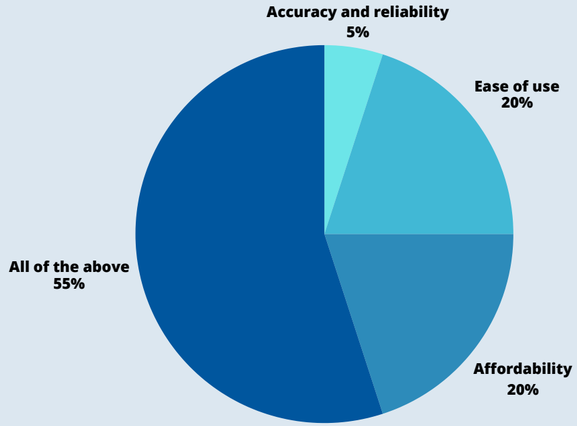
2. How often do you currently visit an ophthalmologist for glaucoma check-ups?



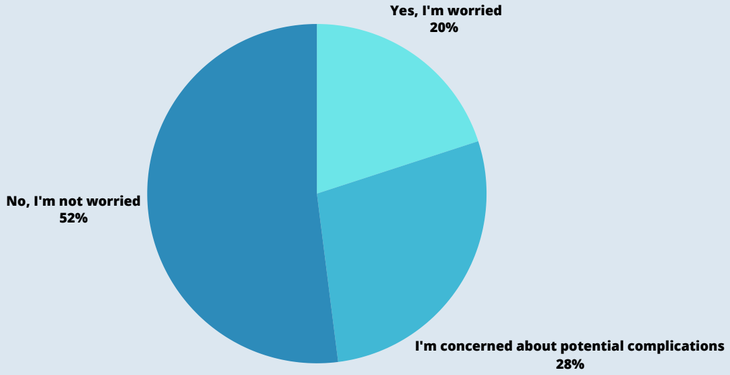
3. Would you prefer a more convenient way to measure your IOP at home?



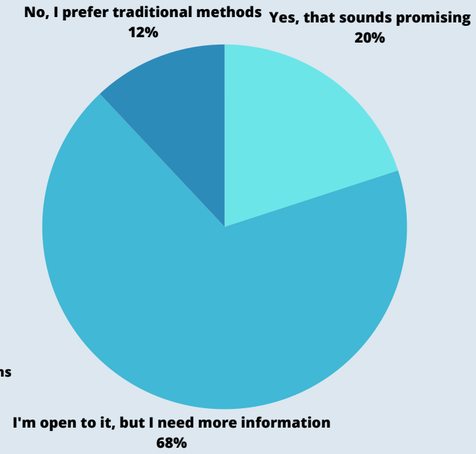
4. What factors are most important to you in a home IOP measurement device?



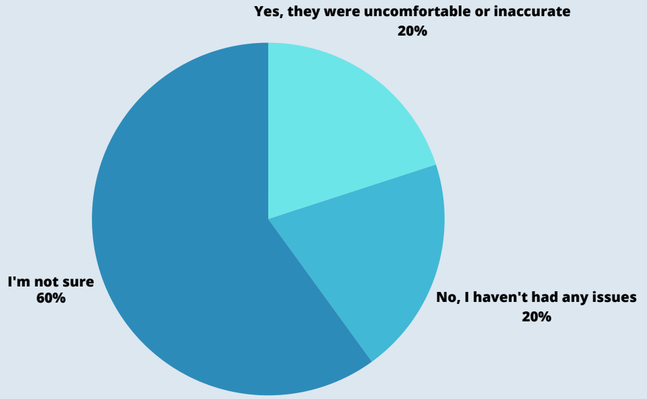
5. Are you concerned about the potential risks or discomfort associated with traditional IOP measurement methods?



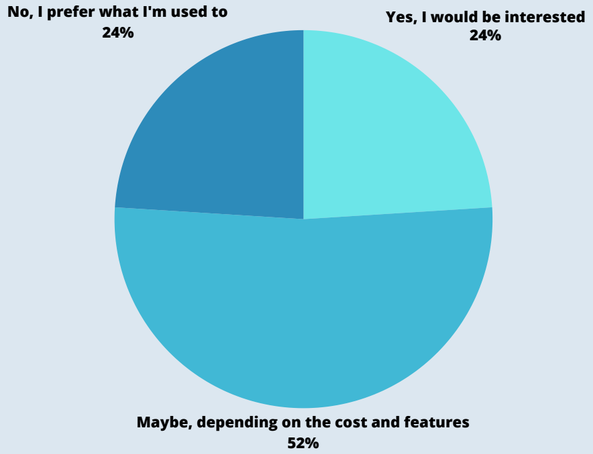
6. Would you be interested in a non-invasive device that uses air puff for IOP measurement?



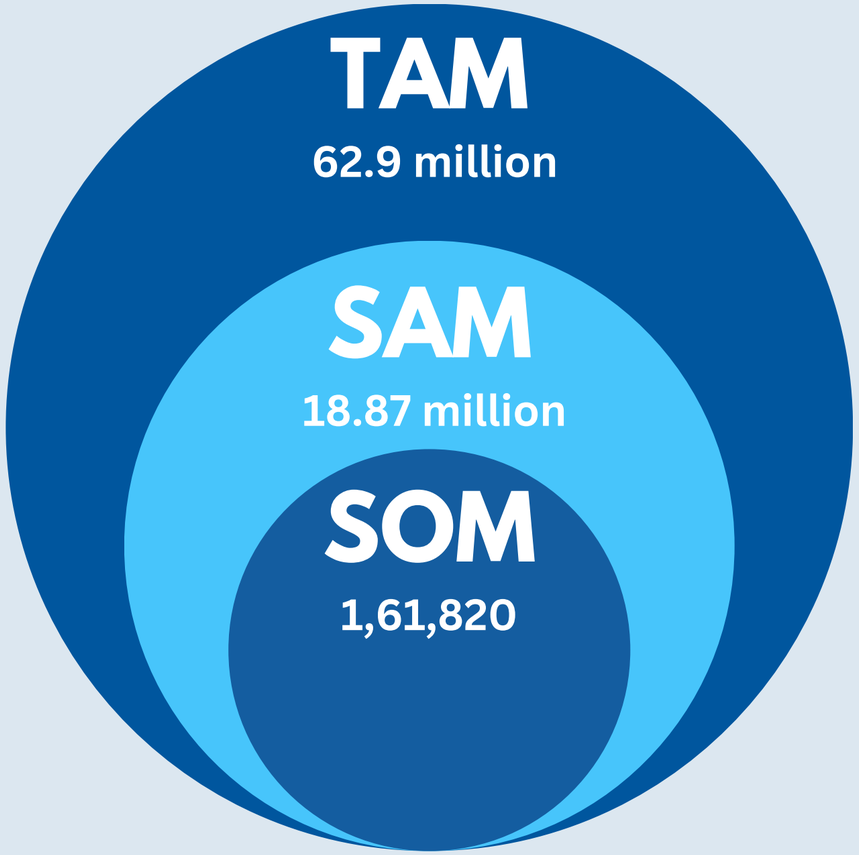
7. Have you ever experienced any difficulties using traditional IOP measurement devices?



8. Would you be willing to try a new IOP measurement device if it was proven to be effective and convenient?



Market size estimation



Market Segment	Description	Number of Customers	Source
TAM (Total Addressable Market)	India: Individuals at risk of glaucoma or requiring regular IOP monitoring.	Approximately 62.9 million people	Glaucoma Prevalence in India
SAM (Serviceable Available Market)	India: Individuals with access to healthcare facilities and technology for regular monitoring. Approximately 50% of TAM.	Estimated 18.87 million (based on factors like geographical reach, pricing, and target customer segments)	Healthcare Access in India
SOM (Serviceable Obtainable Market)	Chennai: Localized market focusing on Chennai residents needing glaucoma monitoring. Assuming Chennai represents ~1% of India's population and similar prevalence.	Estimated 161,820	Population of Chennai

Customer Segment & Niche

Customer Segment	Niche	Description
Individuals	General Population	Adults aged 40 and above, particularly those with a family history of glaucoma or other risk factors.
Individuals	Rural Population	Individuals living in rural areas with limited access to healthcare facilities.
Healthcare Providers	Optometrists and Ophthalmologists	Practitioners seeking a convenient and efficient tool for screening and diagnosing glaucoma.
Healthcare Institutions	Eye Clinics and Hospitals	Institutions looking to improve their glaucoma screening capabilities and patient care.
Government Agencies	Public Health Organizations	Government agencies involved in eye health programs and initiatives.



Customer Persona



Vishal Sharma

Age: 55
Occupation: Retired government employee
Location: Chennai, Tamil Nadu, India

Personality Traits:

Practical

Cautious

Trustworthy

Goals:

- Maintain good vision and independence
- Avoid vision loss and blindness
- Find an affordable and accessible way to monitor glaucoma
- Reduce the burden of frequent doctor visits

Challenges/Frustrations:

- Difficulty in accessing specialized eye care in remote areas
- High cost of glaucoma treatments and medications
- Inconvenience of traditional IOP measurement methods
- Fear of vision loss and its impact on daily life

Bio - Sketch:

Vishal Sharma, 55, is a retired government employee living in Pune. After being diagnosed with early-stage glaucoma, Vishal has to regularly monitor his intraocular pressure (IOP) to prevent vision loss. He has always prioritized his health, but frequent trips to the eye clinic are becoming burdensome.

Though comfortable with smartphones, he prefers simple devices that don't require much technical expertise. He enjoys spending time with his family and values being independent, especially in managing his health. His vision problems have increased his anxiety about potential blindness, and he is keen on finding a less stressful and more convenient way to manage his glaucoma.

Motivations:

Comfort - seeking

Health Conscious

Independent

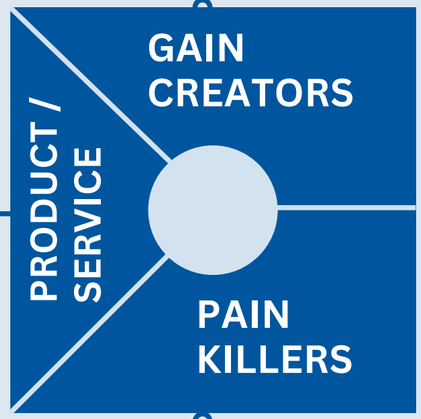
Prevention - oriented

Convenience - driven

Attributes:

- Anxiety-Prone
- Routine-Oriented
- Health-Conscious
- Accessibility-Driven
- Value-Seeker

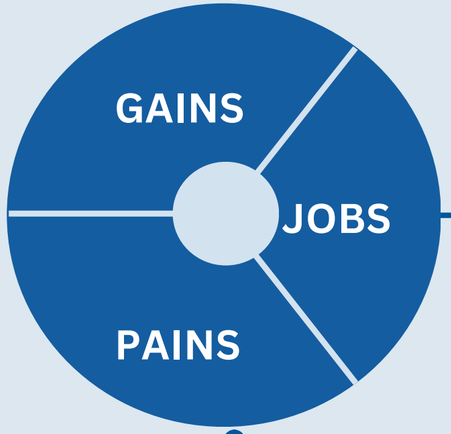
Value proposition Canvas



- Enables early detection of glaucoma, reducing the risk of vision loss.
- Provides accurate and reliable IOP measurements.
- Improves the quality of life for individuals with glaucoma.
- Contributes to public health by reducing the prevalence of glaucoma-related blindness.

- Portable, non-invasive device for measuring IOP using air puff technology.
- Easy-to-use interface and intuitive design for both healthcare professionals and consumers.
- Affordable and accessible pricing.
- Potential for integration with telehealth platforms for remote monitoring.

- Eliminates the need for travel to specialized eye care facilities.
- Provides a comfortable and painless way to measure IOP.
- Reduces the financial burden of glaucoma treatment by enabling early detection and prevention.



- Accurate and reliable glaucoma screening.
- Convenient and accessible IOP measurement.
- Early detection of glaucoma, reducing the risk of vision loss.
- Improved quality of life for individuals with glaucoma.
- Cost savings associated with early detection and prevention.

- Help individuals, especially those in remote or underserved areas, access reliable and affordable glaucoma screening.
- Provide a convenient and comfortable way to measure IOP for individuals at risk of glaucoma.
- Enable early detection of glaucoma to reduce the risk of vision loss.

- Difficulty in accessing reliable and affordable glaucoma screening, especially in remote or underserved areas.
- Inconvenience and discomfort associated with traditional IOP measurement methods.
- Fear of vision loss due to glaucoma.
- Financial burden of glaucoma treatment.

Solution Consists of

Solution	Benefits
Portable Design	Enables easy transport and use in various settings, including homes, clinics, and remote locations.
Non-Invasive Measurement	Eliminates discomfort associated with traditional methods, increasing patient comfort and reducing anxiety.
Air Puff Technology	Provides accurate IOP readings without direct contact, minimizing the risk of infection.
User-Friendly Interface	Simplifies operation for both healthcare professionals and patients, promoting ease of use.
Smart Technology Integration	Allows for real-time data tracking and monitoring, facilitating better disease management and timely interventions.
Telemedicine Compatibility	Enhances access to remote consultations, making it easier for patients to receive care without traveling to a clinic.



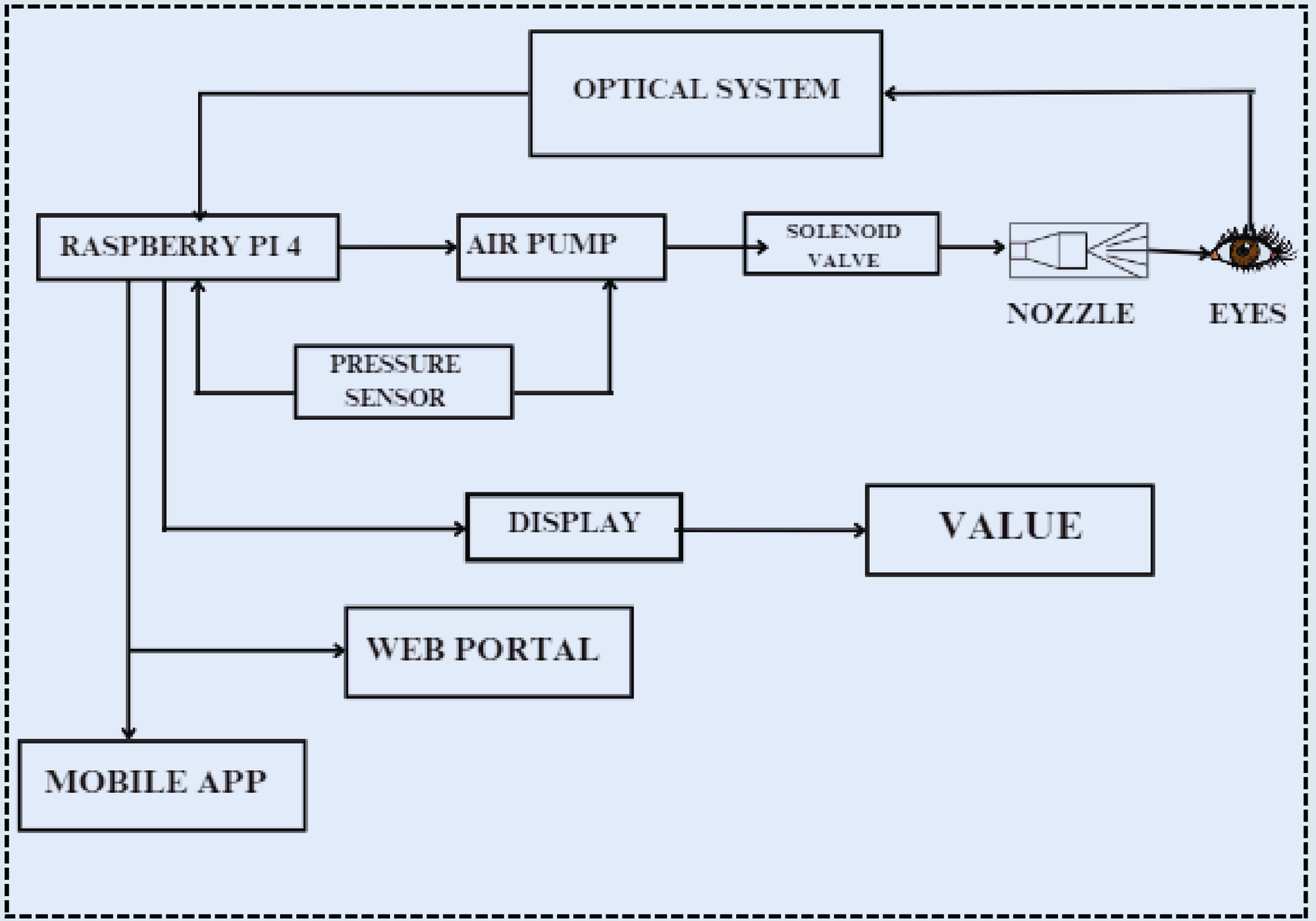




Processes involved

- Image processing
- IOT
- Sensors
- Machine Learning
- Artificial Intelligence
- Data Analysis

Block Diagram



Competition Analysis (Direct Competitors)

Company	Product	UVP	Price (INR)	Market Review	Acceptance	Rating	Website
AirTon	Portable non-invasive glaucoma detection using air puff & interference pattern	Portable, affordable, real-time detection, database creation, non-invasive	Est. ₹20,000–₹30,000	N/A	N/A	N/A	N/A
iCare Home2	Rebound tonometer for self-measurement	First FDA-approved self-use IOP device with a user-friendly interface	₹2,00,000–₹2,50,000	Well-received, widely adopted but challenges with patient training for accuracy	Moderate, but improved with version 2	4.0/5	icare-world.com
Tono-Pen Avia	Handheld electronic tonometer	Lightweight, ergonomic, long battery life, accurate IOP readings	₹1,20,000–₹1,50,000	Popular among clinicians for ease of use, lacks patient-side self-use features	Trusted among professionals but not for home use	4.5/5	ophthalmetryoptical.com
Triggerfish Sensor	Contact lens sensor for 24-hour IOP monitoring	Continuous IOP monitoring via a soft contact lens	₹3,50,000+	Innovative for continuous measurement, but very high cost	Limited due to discomfort and high cost	3.5/5	sensimed.ch
Eyemate-IO	Implantable micro sensor	Implantable, provides real-time wireless IOP measurement	₹4,00,000+	High-end product, accepted for complex cases but invasive and expensive	Low, mostly for surgical patients	3.8/5	implandata.com
Keeler PSL One	Non-contact tonometer	Robust, reliable, optical technology for non-invasive IOP checks	₹2,00,000+	Highly durable and well-regarded in clinics, though expensive for home use	High, but limited to clinics	4.2/5	keeler.co.uk

Competition Analysis (Direct Competitors)

Company	Product Description	UVP	Price (INR)	Market Review & Acceptance	Rating/ Feedback	Website
Icare Finland	Portable iCare tonometer for non-contact IOP measurement.	Handheld, accurate, easy to use without anesthesia	₹1,00,000 - ₹2,00,000	Widely accepted in hospitals globally for its reliability.	4.5/5	www.icarefinland.com
Nidek	NT-510 Non-Contact Tonometer that uses air puff technology.	Reliable air-puff technology with auto-alignment	₹3,00,000 - ₹4,50,000	Popular in ophthalmology clinics; robust and trusted.	4.2/5	www.nidek-intl.com
Reichert Technologies	Tono-Pen handheld digital tonometer.	Compact, affordable, accurate in both supine & sitting positions	₹1,50,000 - ₹3,00,000	High market penetration; widely accepted by professionals.	4.4/5	www.reichert.com
Bionode LLC	Wearable contact lens and glasses combo that modulates IOP.	Non-invasive, uses electromagnetic fields for IOP reduction	₹2,00,000 - ₹3,00,000 (expected)	High interest in clinical trials; potential game-changer.	N/A (prototype)	www.purdue.edu
Diaton	Transpalpebral scleral tonometer (measures through eyelid).	Non-contact, measures without anesthesia through eyelid	₹70,000 - ₹1,00,000	Increasingly recognized in emergency settings and clinics.	4.0/5	www.tonometerdiaton.com

Why Our Venture Can Stand Out:

Affordability: With an estimated price range of ₹20,000–₹30,000, your device is positioned as one of the most affordable options, making it accessible to a broader market compared to competitors like iCare Home2 (₹2,00,000–₹2,50,000) and Triggerfish (₹3,50,000+).

Non-Invasiveness with Unique Technology: While many competitors rely on air puff (Nidek, Reichert), your venture's integration of interference patterns to calculate IOP is an innovative approach that might offer higher accuracy or better patient comfort.

Portability & Accessibility: Similar to Tono-Pen Avia, your device offers portability, but at a significantly lower price, which gives it an edge in accessibility for both clinical and home use.

Data Management: One of the key differentiators is your device's ability to transfer data and create a patient database. This feature can provide valuable insights over time and improve glaucoma management—a feature that is not as advanced in other competing products & a feature many traditional tonometers lack.

Target Market: While competitors often focus on clinical environments or specialized cases, our venture addresses a larger market, including people with high screen time and eye strain, which could lead to early detection and intervention for glaucoma. This approach targets a growing demographic not directly addressed by the other competitors.

Competition Analysis (Indirect Competitors)

Company	Product	Method	Unique Value Proposition (UVP)	Price (INR)	Market Review	Acceptance	Website
Visionix (Luneau)	Visionix VX120+	Multi-diagnostic tool, includes IOP	Combines IOP measurement with other eye tests	~INR 3,00,000+	Highly regarded for comprehensive eye analysis, but high-cost and clinic-based	Moderate to High	Visionix
Topcon	TRK-2P	All-in-one refraction and tonometry device	Offers auto-refraction and IOP measurements	~INR 2,50,000+	Popular in clinics for dual-functionality and ease of use	High	Topcon
Bioptigen (Leica)	Envisu C2300	Optical coherence tomography (OCT) device	Focused on structural imaging, detects eye issues	~INR 5,00,000+	Premium quality imaging for advanced diagnostics, not for self-use	High	Leica
Tomey	FT-1000	Air-puff tonometer	Non-contact, air-puff for IOP measurement	~INR 2,00,000+	Moderate acceptance for standalone tonometry, mostly clinic-based	Moderate	Tomey
Alcon (AcuFocus)	AcuTarget HD	Pupil diagnostics system, indirect IOP support	IOP implications through pupil and light response	~INR 3,50,000+	Mainly adopted by clinics for evaluating IOP-related risk factors	High	Alcon
EyeQue	Personal Vision Tracker	Smartphone-based refraction device	Low-cost, self-use vision tester for eye health	~INR 15,000	Gaining popularity among consumers for accessibility, but not IOP-specific	Moderate	EyeQue
Glaukos	iStent inject W	Surgical implant for pressure control	Long-term IOP reduction, but requires surgery	Not listed; high	Used for chronic cases in clinics; invasive but effective for severe glaucoma	High (chronic cases)	Glaukos

Business Model Canvas

Tagline : "Where Technology Meets Life for Better Health."

DESIGNED FOR:	DESIGNED BY:	DATE:	VERSION:
Eye care patients, hospitals, clinics, and home healthcare providers	SARAVANAN R K & Team AirTon	28.04.2025	1.0

KEY PARTNERS

- Leading eye hospitals and ophthalmology clinics
- Medical device manufacturing firms
- Regulatory bodies (CDSCO, CE, ISO)
- Insurance companies and healthcare financing partners
- Telemedicine platforms
- Non-profits and rural health initiatives
- University research labs and innovation hubs

KEY ACTIVITIES

- High-precision device R&D and prototyping
- Clinical validation and pilot testing
- Compliance with medical regulations
- Scalable and cost-effective manufacturing
- Strategic marketing and thought leadership
- Post-sales support and patient engagement

KEY RESOURCES

- Multidisciplinary R&D team (hardware, software, clinical)
- Patented air-puff technology + IOP monitoring algorithms
- State-of-the-art labs and manufacturing units
- Strong network with ophthalmologists and healthcare providers
- Financial backing (grants, VC, CSR initiatives)

VALUE PROPOSITIONS

- First-in-class portable dual eye simultaneous IOP monitoring with real-time data
- Non-invasive, safe, and patient-friendly
- Dual-eye simultaneous scanning for faster diagnosis
- Affordable pricing, 10x cheaper than hospital tonometers
- Smart alerts for early detection to prevent vision loss
- Future-ready: Integrated with cloud-based health monitoring
- Empowering rural and remote healthcare accessibility

CUSTOMER RELATIONSHIP

- Patient-centric onboarding and training
- Personalized device setup and education
- Loyalty programs for clinics and doctors
- In-app reminders and AI-driven health tips (future)
- Dedicated customer support helpline and service centers

CHANNEL

- Direct sales via hospitals and eye care chains
- Strategic partnerships with telemedicine platforms
- E-commerce platforms (Amazon Healthcare, Flipkart Health+)
- Presence in medical exhibitions and conferences (e.g., AIOS)
- Government tenders and rural health outreach
- Influencer marketing via leading ophthalmologists

CUSTOMER SEGMENTS

- Glaucoma patients
- Eye care professionals,
- Clinics
- Hospitals
- Telemedicine platforms
- Remote or underserved populations

COST STRUCTURE

- Intensive R&D and clinical trials
- Regulatory certifications (CDSCO, CE)
- Manufacturing, quality control, and logistics
- Branding, marketing, and salesforce management
- Customer support and warranty services
- App and software development (future upgrade)

REVENUE STREAMS

- One-time device sales to individuals and clinics
- Customized hospital packages and bulk orders
- Extended warranty and annual maintenance contracts
- SaaS model: Subscription for cloud health analytics (future)
- Licensing patented technology to global partners

MVP Description

Feature	Description
Non-Invasive IOP Measurement	Uses air puff technology to measure intraocular pressure (IOP) without physical contact, ensuring comfort.
Interference Pattern Technology	Incorporates interference patterns for precise and accurate IOP calculations.
Portable and Lightweight	Compact, battery-powered device designed for ease of use in clinics, homes, and mobile screening units.
Affordable Solution	Priced under ₹50,000 to ensure accessibility, especially for rural and underserved healthcare centers.
Data Transfer and Integration	Enables digital data transfer for remote monitoring and integration with telemedicine platforms.
User-Friendly Interface	Intuitive interface that allows easy operation by both healthcare professionals and patients.
Real-Time Data Analysis	Provides immediate results and creates a patient database for long-term monitoring and trend analysis.
Battery Life	Long-lasting battery suitable for multiple tests in a single charge, ideal for mobile use.
Safe and Hygienic	Completely non-contact method, reducing the risk of infection and eliminating the need for anesthesia.
Target Market Focus	Designed to cater to individuals with high screen time, those at high risk of glaucoma, and regular patients.

MVP



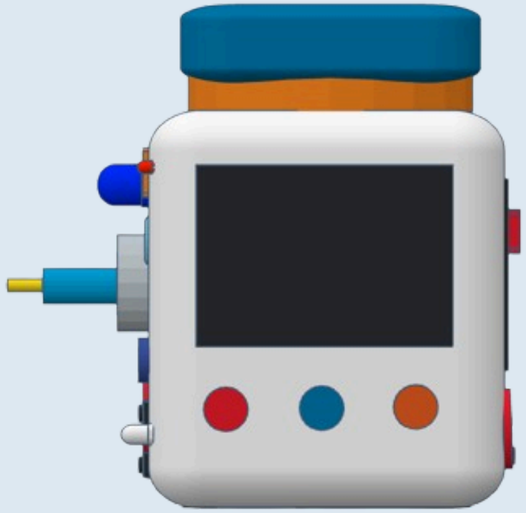
Front view



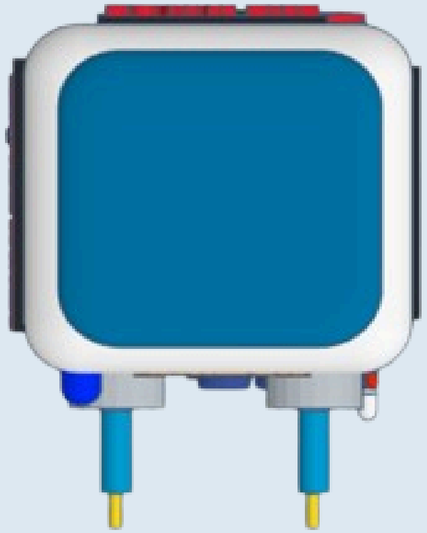
Left Side view



Back view



Right Side view



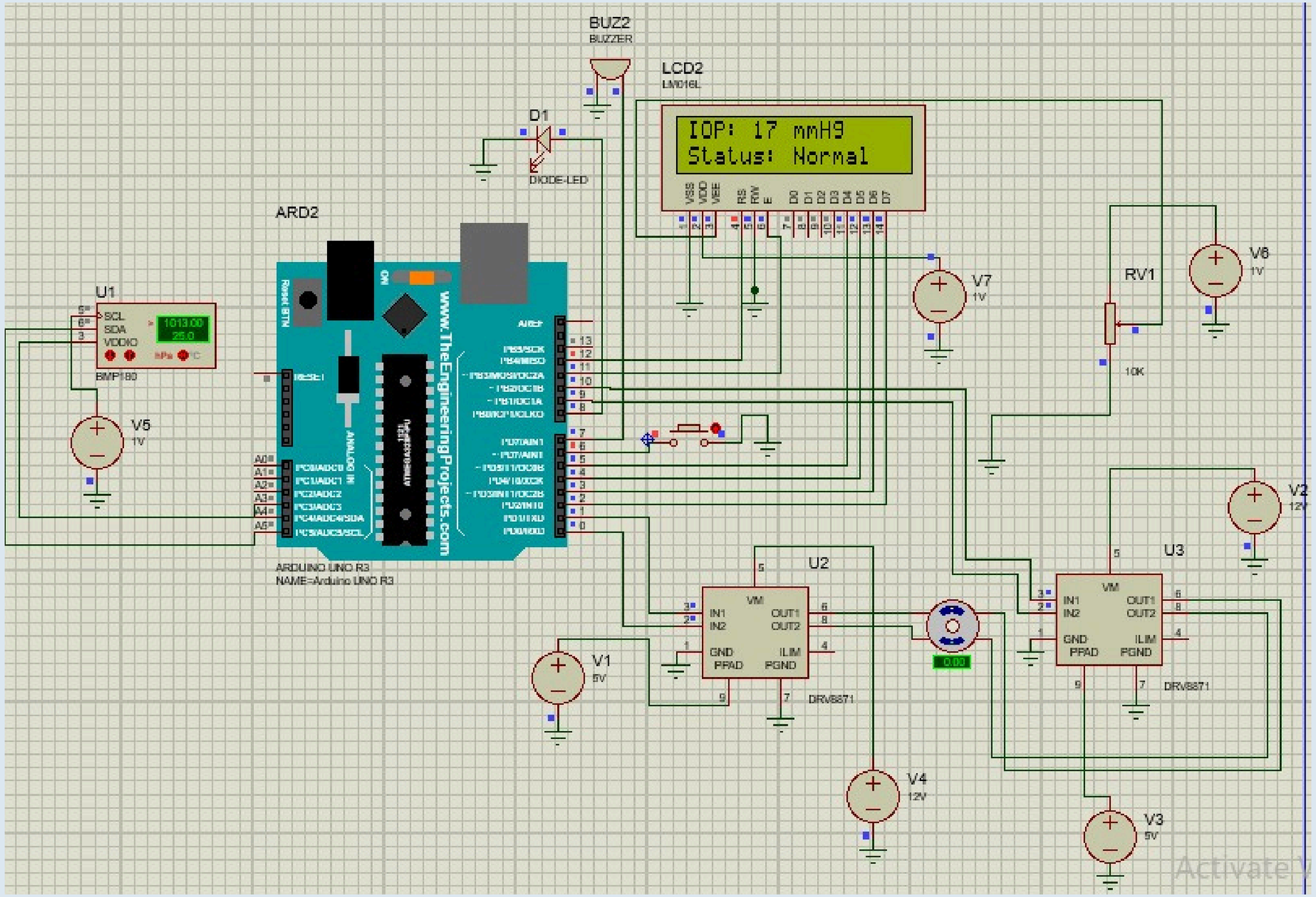
Top view

Our MVP is a dual-eye, portable, non-invasive device designed to measure intraocular pressure (IOP) simultaneously in both eyes using air puff technology. This innovative approach enhances efficiency, accuracy, and patient comfort, addressing key limitations of existing single-eye tonometers.

Key Features of the MVP:

- Dual-Eye IOP Measurement: Simultaneous assessment for improved accuracy and reduced testing time.
- Portable & Non-Invasive Design: Compact and easy to use, enabling monitoring outside clinical settings.
- Air Puff Technology: Ensures a quick, painless, and contact-free measurement.
- Basic User Interface: Simple controls for ease of use by medical professionals and potential home users.
- Prototype Validation: Focused on testing measurement accuracy, consistency, and reliability.

Simlation via Proteus

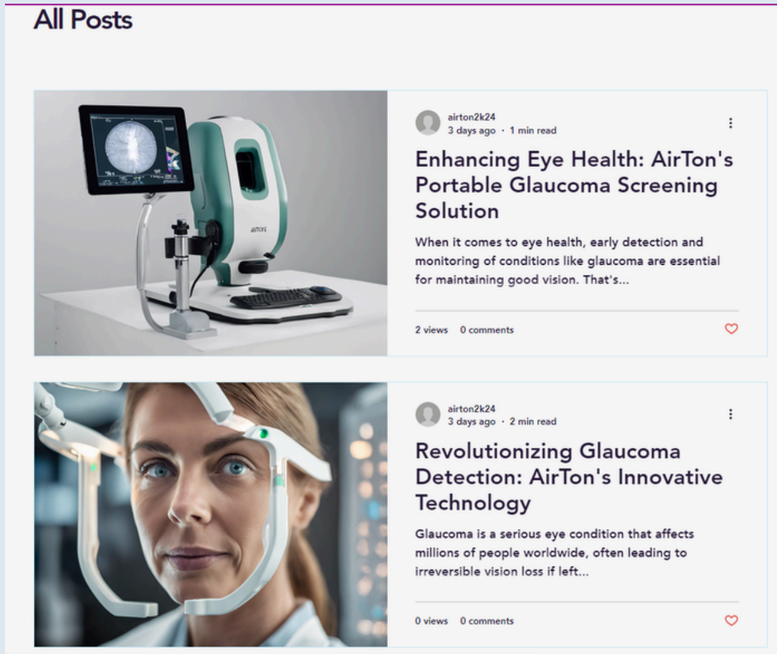


MVP (Website)



About us

Our venture aims to address the challenges faced by patients by developing a portable, non-invasive device that utilizes air puff technology to measure IOP. This solution will offer a more patient-friendly approach, reducing discomfort while enabling frequent monitoring outside of clinical environments. By making glaucoma detection and IOP measurement more accessible, our device has the potential to revolutionize early diagnosis and long-term management of the disease, ultimately reducing the global burden of blindness caused by glaucoma.



Our Vision

"To empower individuals worldwide with accessible and innovative eye care solutions, enabling early detection and prevention of glaucoma, and ensuring that everyone has the opportunity to maintain clear vision and optimal eye health."



Website: https://airton2k24.wixsite.com/airton

MVP (Mobile version)

AIRTON

Search...



About us

Our venture aims to address the challenges faced by patients by developing a portable, non-invasive device that utilizes air puff technology to measure IOP. This solution will offer a more patient-friendly approach, reducing discomfort while enabling frequent monitoring outside of clinical environments. By making glaucoma detection and IOP measurement more accessible, our device has the potential to revolutionize early diagnosis and long-term management of the disease, ultimately reducing the global

Our Vision

"To empower individuals worldwide with accessible and innovative eye care solutions, enabling early detection and prevention of glaucoma, and ensuring that everyone has the opportunity to maintain clear vision and optimal eye health."



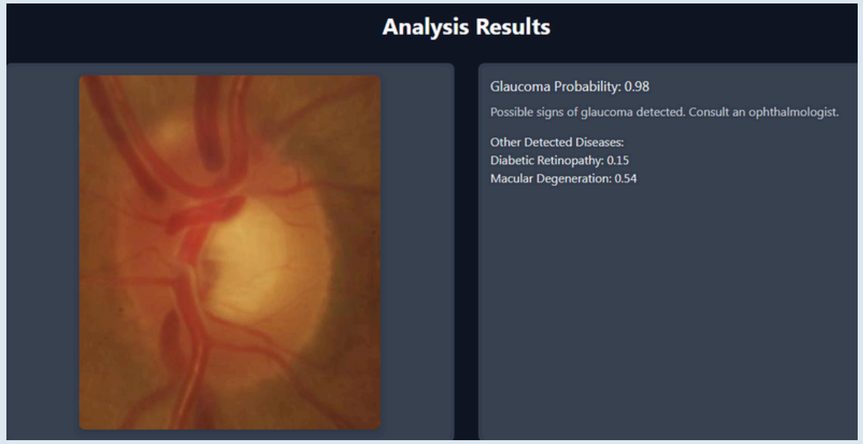
AirTon

airton2k24.wixsite.com
airton2k24@proton.me

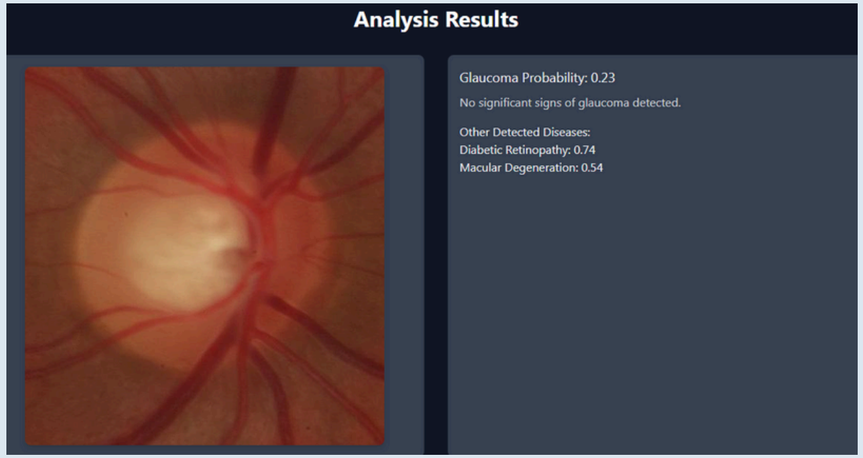
RESEARCH AND REFERENCES

REFERENCES

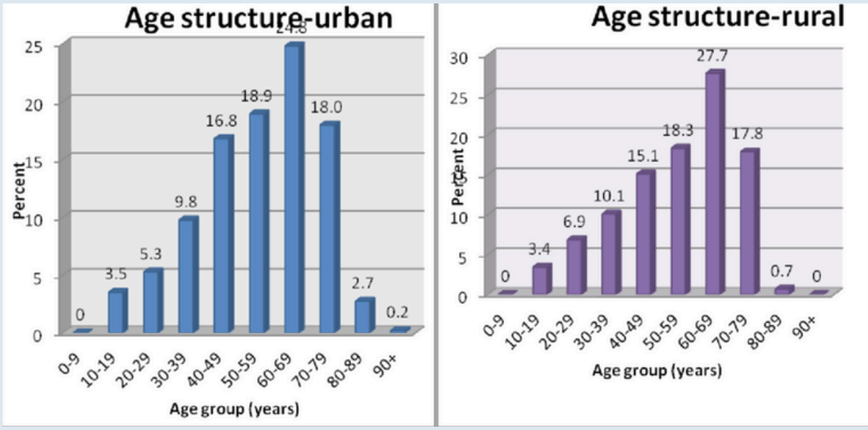
1. B. K. Koirala, R. Shrestha, and S. Shrestha, "On imaging based non-contact tonometer for intraocular pressure measurement," 2012 IEEE International Conference on Imaging Systems and Techniques Proceedings, Manchester, UK, 2012, pp. 618-622.
2. R. K. Kodali and S. S. Mandal, "Intraocular Pressure Monitoring Using IoT," 2022 IEEE Global Conference on Computing, Power and Communication Technologies (GlobConPT), Thiruvananthapuram, India, 2022, pp. 1-4.
3. M. Křížek, H. Kaneko, and E. Sugimoto, "Understanding eye deformation in non-contact tonometry," 2008 30th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, Vancouver, BC, Canada, 2008, pp. 923-926.
4. S. Chinnaswamy, V. Natarajan, and S. Samiappan, "Intraocular Pressure Monitoring System for Glaucoma Patients Using IoT and Machine Learning," Engineering Proceedings, vol. 59, no. 1, p. 179, 2023.



GLAUCOMA DETECTED EYE



NON - GLAUCOMA EYE



PATIENTS AFFECTED BY GLAUCOMA AMONG DIFFERENT AGE GROUPS IN RURAL AND URBAN AREAS

OUTPUT

https://aravinds2006-airton-node.hf.space/

GUIDED BY:

Dr. J. Thamil Selvi , Head of the Department, Department of Electronics and Communication Engineering, Sri Sairam Engineering College, Chennai

MVP Validation

- How many customers validated your MVP? - 25
- What was the Validation mode? - Google Forms
- How many of them were satisfied with your MVP? - 12
- How many of them said that MVP needs some improvement? - 9
- How many of them said that they were not satisfied? - 4

Google forms link:

https://docs.google.com/forms/u/1/d/e/1FAIpQLScmQ8suWO0Elpp159Lo7s2Sjo_RJW21lfVqNE-TkAYTSj07PQ/formResponse

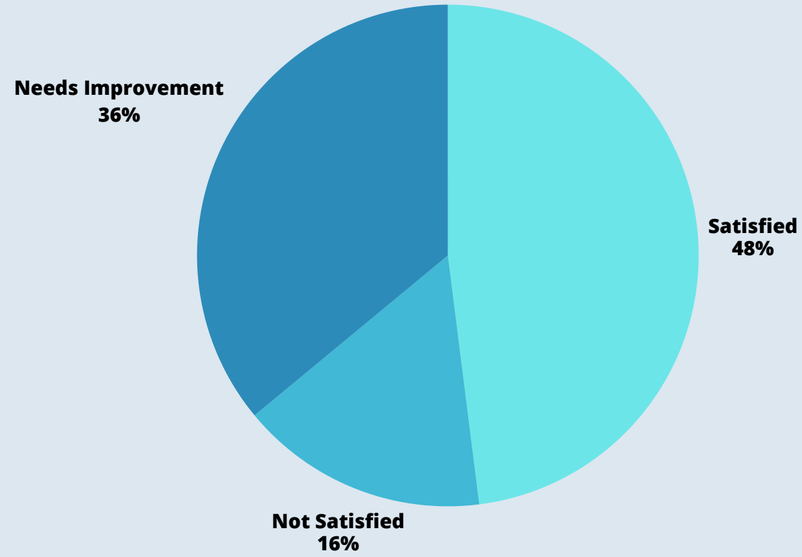
What is your MVP ? : 3D model

Who is our target audience for the test? How many of them?

Primary Audience: Ophthalmologists, optometrists, and glaucoma patients.
Secondary Audience: Individuals with high screen time, people with other eye conditions, and people at risk for glaucoma.
Sample Size: You should target testing on 20-30 patients across various demographics and eye conditions to get comprehensive feedback and validation.

How will we get to that audience?

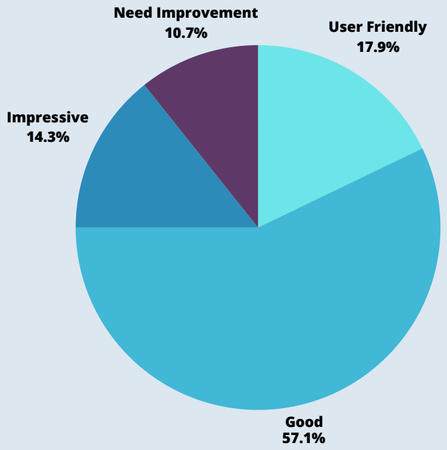
Partnership with Clinics and Hospitals: Collaborate with ophthalmology departments and clinics to recruit glaucoma patients for testing.
Social Media Outreach: Targeted campaigns on social media platforms like Facebook, Twitter, and Instagram to reach individuals concerned about eye health.
Support Groups and Organizations: Partner with glaucoma support groups and healthcare organizations to find participants who are at risk or already undergoing treatment.
Direct Referral: Optometrists and eye care professionals can recommend patients to participate in the MVP validation.



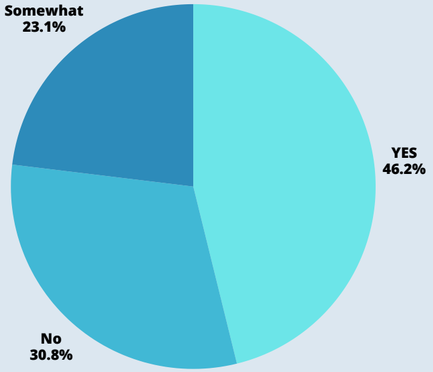
Realization & Insights from the Analysis

- Many users appreciate the non-invasive nature of the device but highlight that ease of use is critical. Complex operation or unclear instructions can deter adoption.
- Users are concerned about comfort, particularly regarding the air puff mechanism. This realization emphasizes the need for a more user-friendly design to enhance comfort and reduce anxiety.
- Interest in data transfer and smartphone integration shows a market demand for connected health devices. However, technical issues in this area suggest that functionality needs to be refined.Variability in measurements indicates a potential issue with consistency. Users' concerns highlight the need for calibration and guidance on optimal usage conditions to ensure accurate readings.

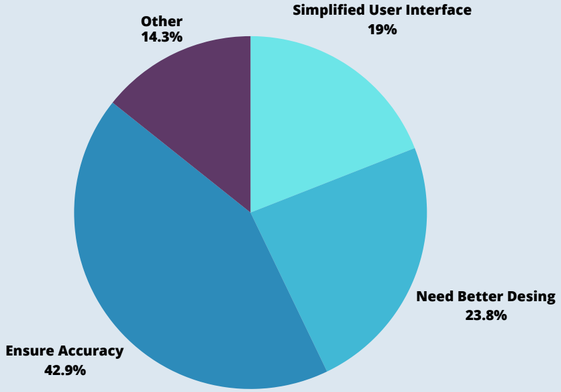
1.What are your first impressions of the 3D model of the device?



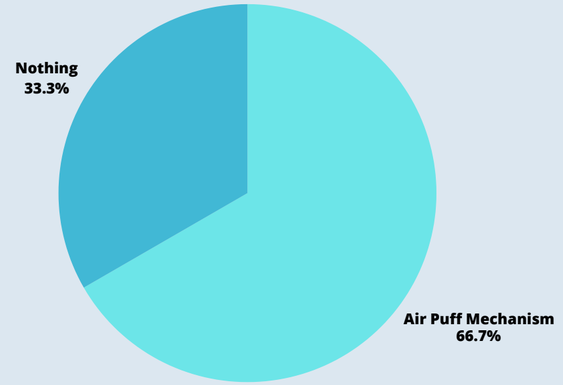
2.Do you find the design of the device appealing and user-friendly?



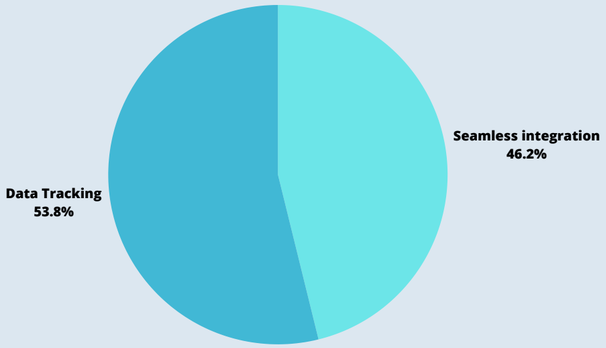
3.What improvements or additional features would you suggest for the device?



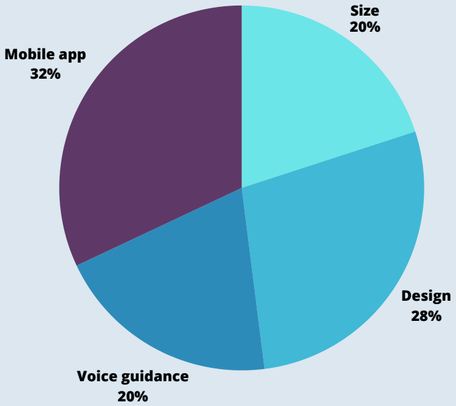
4.Are there any aspects of the design that you find confusing or difficult to understand?



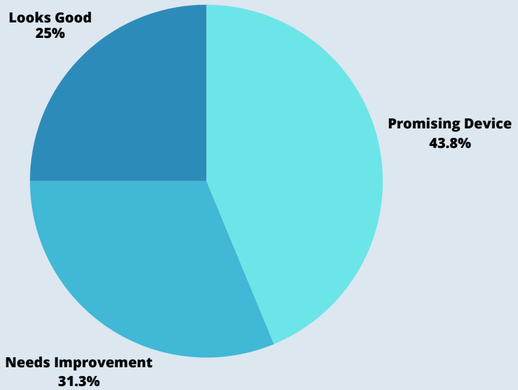
5. What other features or improvements would you like to see in future versions of the device?



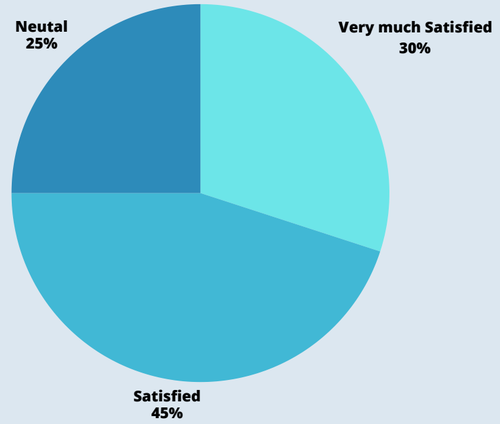
6.If you could improve one aspect of the device, what would it be?



7.Do you have any additional comments or suggestions about the device?



8.Overall, how satisfied are you with the device’s performance? Would you recommend it to others who are concerned about glaucoma or IOP monitoring?



Go-To-Market Strategy

GTM Strategy Component	Details
Market Research & Analysis	**Target Market:** Individuals at risk of glaucoma (aged 40+), people with high screen time, and those with existing eye problems. **Geographic Focus:** Urban areas initially, rural expansion later. **Competitive Analysis:** Research competitors, positioning device as affordable, portable, and easy-to-use compared to hospital-grade devices.
Product Positioning	**Unique Selling Proposition (USP):** Non-invasive, affordable, portable device with air puff technology for accurate IOP measurement. **Differentiation:** Faster, more affordable, and easier to use than traditional glaucoma detection methods. Self-monitoring for home use, IoT integration for remote doctor consultation.
Pricing Strategy	**Basic Model:** ₹25,000 with standard features for home users. **Advanced Model:** ₹40,000 (with IoT connectivity and advanced sensors). **Discounts:** Seasonal offers, bulk discounts for clinics, subscription plans for remote monitoring. **Flexible Payments:** EMI options.
Sales & Distribution Channels	**Online Sales:** E-commerce website, Amazon, Flipkart. **Offline Sales:** Partnerships with ophthalmology clinics, optometrists, hospitals. **Direct-to-Consumer:** Promote via social media, targeted ads, and influencers.

Go-To-Market Strategy

GTM Strategy Component	Details
Marketing & Promotion	**SEO:** Optimize website for keywords related to glaucoma, IOP measurement, and home care. **Social Media:** Facebook, Instagram, LinkedIn for product demos and customer testimonials. **Influencer Marketing:** Partner with health influencers and ophthalmologists. **Paid Ads:** Use Google Ads, Instagram Ads for direct outreach. **Referral Program:** Offer discounts for customer referrals.
Partnerships & Collaborations	**Healthcare Providers:** Partner with eye care professionals to recommend the device. **NGOs/Government:** Collaborate with health organizations for outreach. **Insurance:** Work with insurance companies to include the device in health plans for glaucoma patients.
Customer Support & After-Sales	**Technical Support:** Live chat, phone, and email support. **Warranty:** 1-2 years for repairs/replacements. **Remote Monitoring:** Collaborate with doctors for remote monitoring of IOP and offer consultation services.
Timeline & Execution Plan	**Phase 1 (Months 1-3):** Finalize product and regulatory approvals, website and SEO setup, social media presence. **Phase 2 (Months 4-6):** Product launch, feedback gathering, clinic distribution, influencer marketing. **Phase 3 (Months 7-12):** Scale marketing, increase distribution, build partnerships with hospitals, eye clinics, and insurance companies.
Metrics & KPIs	**Sales Metrics:** Units sold, revenue, average order value. **Customer Acquisition:** Cost per acquisition, conversion rates. **Customer Retention:** Repeat purchases, satisfaction ratings, referral rates. **Market Penetration:** Sales by region, market share growth.

Sales Funnel



*This Sales Funnel is for One Year

Stage	Description	Assumption	Percentage of SOM	In Numbers
Target Market (SOM)	Individuals at risk of glaucoma, including high screen-time users and those with eye health issues.	Approximately 10% of a regional population, focusing on accessible urban areas.	100%	1.62 Lakhs
Leads (Months 1-3)	Initial interest from health campaigns, ads, and awareness content.	25% of the SOM expresses interest, attracted by convenience and non-invasiveness.	25%	40,500
Prospects (Months 4-9)	Leads actively considering the device through demos, trials, or health expos.	40% of leads progress to this stage after evaluating the device's value.	10%	16,200
Customers (Months 10-12)	Final buyers who see value in self-monitoring and early detection.	30% of prospects convert to customers, driven by effectiveness, affordability, and testimonials.	3%	4,860

Customer Acquisition Plan

Stage	Objective	Target Market	Strategies	Metrics for Success
Market Research & Segmentation	Identify specific customer segments within target markets	Ophthalmologists, Optometrists, Telemedicine Providers, Clinics, Hospitals, High-Risk Individuals (e.g., high screen time users, diabetics)	Surveys, Focus Groups, and Analysis of Local & National Data; Competitor Analysis	Defined customer personas; insights into needs
Awareness Campaign	Build brand awareness	General Public, Healthcare Professionals, Clinics, Government Health Departments	Digital Marketing (Social Media Ads, SEO, Blogs); Industry Events and Webinars; Influencer Partnerships (Ophthalmologists, Eye Health Advocates)	Engagement rates; Reach and impressions on ads
Lead Generation	Drive interest and collect leads	Eye Care Providers, Clinics, Private Healthcare Centers	Landing Page with Educational Content; Newsletter Signup; Free Educational Webinars on Glaucoma	Number of leads; Conversion rate on landing page
Outreach & Direct Sales	Personalized engagement with high-priority leads	Clinics, Ophthalmologists, Optometry Chains	Direct Outreach (emails, calls, and demos); Partnerships with Optometry Associations	Number of meetings/demo requests; Response rate
Free Trials & Pilots	Build trust through firsthand experience	Private Clinics, Hospitals, High-Risk Patients	Limited-time free trials or pilot program; Testimonials and case studies from pilot participants	Trial signups; Feedback quality; Conversion rate post-trial

Customer Acquisition Plan

Stage	Objective	Target Market	Strategies	Metrics for Success
Content Marketing	Educate market on benefits of non-invasive glaucoma screening	Ophthalmologists, Patients at Risk, Clinics, Telemedicine Platforms	Blog Articles, Video Content (demos, expert interviews); Glaucoma Awareness Campaigns (e.g., during World Glaucoma Week)	Increase in website traffic; Time spent on page
Partnership Development	Expand reach through strategic partnerships	Telemedicine Companies, Eye Health Organizations, NGOs	Collaborate with telemedicine platforms, NGOs for rural healthcare, and eye care organizations	Partnership agreements; Co-marketing initiatives
Testimonials & Case Studies	Establish credibility through real-world success stories	Healthcare Professionals, Patients	Gather testimonials and success stories from early adopters; Create case studies on outcomes and device benefits	Number of testimonials; Case study views/downloads
Online and Offline Events	Reach potential customers directly	Eye Health Conferences, Healthcare Seminars	Participate in eye health expos, medical conferences, and healthcare trade shows; host Q&A sessions at events	Leads generated from events; Event engagement
Retargeting & Follow-Up	Maintain engagement with warm leads	All initial lead types and website visitors	Retargeting Ads, Email Follow-Ups with personalized content based on initial engagement	Re-engagement rate; Email open/click-through rates

Start up Costs

Startup Cost	**₹8,60,000**
Business Registration Fees	₹10,000
Ophthalmic Device R&D (Initial Prototype)	₹7,20,000
Website Design and Development	₹30,000
Basic Computer Systems for Data Processing	₹1,00,000
Capital Work in Progress (Fixed Asset)	**₹1,90,000**
Air-Puff Mechanism and IOP Sensor Development	₹1,15,000
High-Resolution Mini-Cameras for IOP Detection	₹75,000
Starting Operations (Budgeted)	**₹2,50,000**
Product Launch Materials (Digital and Print)	₹50,000
Rental Expenses for Lab/Testing Spaces	₹1,00,000
Regulatory Documentation and Compliance Preparation	₹50,000
Stationery, Office Supplies, and Brochures	₹10,000
Miscellaneous	₹40,000
Start-up capital	**₹13,00,000**
Share Captial	₹3,50,000
Loans (from Bank or Other Sources)	₹9,50,000

- The total investment will be INR 13 Lakhs.
- The product is offered for Rs. 25,000 to Rs.35,000 per piece.

- The startup costs are allocated INR 8,60,000, which includes business registration fees of INR 10,000, R&D for developing the initial device prototype at INR 7,20,000, website design and development at INR 30,000, and a basic computer system for development at INR 1,00,000.

- For capital work in progress (fixed assets), a total of INR 1,90,000 has been set aside. This includes INR 1,15,000 for setting up the air-puff mechanism and INR 75,000 for acquiring mini-cameras and sensor components, which are essential for accurate intraocular pressure (IOP) measurements.

- The starting operations budget of INR 2,50,000 will cover product launch expenses (INR 50,000), rental expenses for lab space (INR 1,00,000), regulatory documentation (INR 50,000), and miscellaneous office supplies (INR 10,000) to ensure smooth functioning.

- The startup capital will be sourced from INR 3,50,000 in share capital provided by the team and an additional INR 9,50,000 in loans, bringing the total startup capital to INR 6,00,000.

Financial Statement: https://docs.google.com/spreadsheets/d/1hx_keC0B2tl4alcs2rqSxzl5p275wOrlFDWPtsfTUQ8/edit?usp=sharing

Forecast Profit & Loss

Financial Statement: https://docs.google.com/spreadsheets/d/1hx_keC0B2tl4alcs2rqSxzl5p275wOrlFDWPtsfTUQ8/edit?usp=sharing

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Totals
Sales													
Cash Sales	₹ 1,75,000	₹ 2,80,000	₹ 3,50,000	₹ 4,55,000	₹ 5,95,000	₹ 6,65,000	₹ 5,95,000	₹ 7,00,000	₹ 5,60,000	₹ 5,95,000	₹ 6,30,000	₹ 7,00,000	₹ 63,00,000
No. of Customers	5	8	10	13	17	19	17	20	16	17	18	20	180
Price Per Unit	₹ 35,000	₹ 35,000	₹ 35,000	₹ 35,000	₹ 35,000	₹ 35,000	₹ 35,000	₹ 35,000	₹ 35,000	₹ 35,000	₹ 35,000	₹ 35,000	₹ 36,333
Other Cash Receipts (Consulting Services)	₹ 20,000	₹ 20,000	₹ 20,000	₹ 20,000	₹ 20,000	₹ 20,000	₹ 20,000	₹ 20,000	₹ 20,000	₹ 20,000	₹ 20,000	₹ 20,000	
Total Sales	₹ 1,95,000	₹ 3,00,000	₹ 3,70,000	₹ 4,75,000	₹ 6,15,000	₹ 6,85,000	₹ 6,15,000	₹ 7,20,000	₹ 5,80,000	₹ 6,15,000	₹ 6,50,000	₹ 7,20,000	₹ 65,40,000
COGS													
Cost of Services	₹ 1,25,000	₹ 2,00,000	₹ 2,50,000	₹ 3,25,000	₹ 4,25,000	₹ 4,75,000	₹ 4,25,000	₹ 5,00,000	₹ 4,00,000	₹ 4,25,000	₹ 4,50,000	₹ 5,00,000	₹ 45,00,000
Cost Per Unit	₹ 25,000	₹ 25,000	₹ 25,000	₹ 25,000	₹ 25,000	₹ 25,000	₹ 25,000	₹ 25,000	₹ 25,000	₹ 25,000	₹ 25,000	₹ 25,000	₹ 25,000
Gross profit	₹ 50,000	₹ 80,000	₹ 1,00,000	₹ 1,30,000	₹ 1,70,000	₹ 1,90,000	₹ 1,70,000	₹ 2,00,000	₹ 1,60,000	₹ 1,70,000	₹ 1,80,000	₹ 2,00,000	₹ 18,00,000
Fixed Expenses													
Salaries	₹ -	₹ -	₹ -	₹ 1,08,000	₹ 1,08,000	₹ 1,08,000	₹ 1,08,000	₹ 1,08,000	₹ 1,08,000	₹ 1,08,000	₹ 1,08,000	₹ 1,08,000	₹ 9,72,000
Marketing and Promotion	₹ 4,000	₹ 4,000	₹ 4,000	₹ 4,000	₹ 4,000	₹ 4,000	₹ 4,000	₹ 4,000	₹ 4,000	₹ 4,000	₹ 4,000	₹ 4,000	₹ 48,000
Digital Marketing	₹ 3,000	₹ 3,000	₹ 3,000	₹ 3,000	₹ 3,000	₹ 3,000	₹ 3,000	₹ 3,000	₹ 3,000	₹ 3,000	₹ 3,000	₹ 3,000	₹ 36,000
Utilities (Electricity etc.)	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 24,000
Customer Service	₹ 4,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 59,000
Rent	₹ 10,000	₹ 10,000	₹ 10,000	₹ 10,000	₹ 10,000	₹ 10,000	₹ 10,000	₹ 10,000	₹ 10,000	₹ 10,000	₹ 10,000	₹ 10,000	₹ 1,20,000
Total Expenses	₹ 23,000	₹ 24,000	₹ 24,000	₹ 1,32,000	₹ 1,32,000	₹ 1,32,000	₹ 1,32,000	₹ 1,32,000	₹ 1,32,000	₹ 1,32,000	₹ 1,32,000	₹ 1,32,000	₹ 12,59,000
Result													
Net Profit/Loss	₹ 27,000	₹ 56,000	₹ 76,000	₹ -2,000	₹ 38,000	₹ 58,000	₹ 38,000	₹ 68,000	₹ 28,000	₹ 38,000	₹ 48,000	₹ 68,000	₹ 5,41,000
Gross Profit Margin	29%	29%	29%	29%	29%	29%	29%	29%	29%	29%	29%	29%	29%
Net Profit Margin	15%	20%	22%	0%	6%	9%	6%	10%	5%	6%	8%	10%	9%

Funding Plan & Projections

Financial Statement: https://docs.google.com/spreadsheets/d/1hx_keC0B2tl4alcs2rqSxzl5p275wOrIFDWPtsfTUQ8/edit?usp=sharing

	Year 1	Year 2	Year 3	Year 4	Year 5
Number of sales	180	222	266	324	348
Value of each sale (Contribution)	₹ 11,333	₹ 11,333	₹ 15,028	₹ 15,028	₹ 18,659
Average Price per unit	₹ 36,333	₹ 36,333	₹ 43,600	₹ 43,600	₹ 47,232
Average Cost per unit	₹ 25,000	₹ 25,000	₹ 28,573	₹ 28,573	₹ 28,573
Total revenue	₹ 65,40,000	₹ 80,66,000	₹ 1,16,15,040	₹ 1,41,12,448	₹ 1,64,56,343
Gross Profit	₹ 20,40,000	₹ 25,16,000	₹ 40,03,326	₹ 48,64,101	₹ 65,01,227
Capital costs					
Land and building	N.A.	N.A.	N.A.	N.A.	N.A.
Equipment	₹ 1,50,000	₹ 1,50,000	₹ 1,50,000	₹ 1,50,000	₹ 1,50,000
Product development costs	₹ 1,90,000	₹ 1,90,000	₹ 1,90,000	₹ 1,90,000	₹ 1,90,000
Others	₹ 15,000	₹ 15,000	₹ 15,000	₹ 15,000	₹ 15,000
Total	₹ 3,55,000	₹ 3,55,000	₹ 3,55,000	₹ 3,55,000	₹ 3,55,000
Expenses (Annual)					
Salaries	₹ 9,72,000	₹ 10,69,200	₹ 11,76,120	₹ 12,93,732	₹ 14,23,105
Marketing and Promotion	₹ 48,000	₹ 52,800	₹ 58,080	₹ 63,888	₹ 70,277
Digital Marketing & Customer Services	₹ 36,000	₹ 39,600	₹ 43,560	₹ 47,916	₹ 52,708
Utilities (electricity etc.)	₹ 24,000	₹ 26,400	₹ 29,040	₹ 31,944	₹ 35,138
Office Supplies	₹ 59,000	₹ 64,900	₹ 71,390	₹ 78,529	₹ 86,382
Rent	₹ 1,20,000	₹ 1,32,000	₹ 1,45,200	₹ 1,59,720	₹ 1,75,692
Total	₹ 12,59,000	₹ 13,84,900	₹ 15,23,390	₹ 16,75,729	₹ 18,43,302
Earnings (EBITDA)	₹ 7,81,000	₹ 11,31,100	₹ 24,79,936	₹ 31,88,372	₹ 46,57,925

Note: A 20% YoY linear growth in sales is projected.

Break-even analysis

Financial Statement: https://docs.google.com/spreadsheets/d/1hx_keC0B2tl4alcs2rqSxzl5p275wOrlFDWPtsfTUQ8/edit?usp=sharing

Break-even analysis	
Average sales price per unit	₹ 35,000
Average cost of each unit	₹ 25,000
Gross Profit Margin	29%
Fixed costs for the year	₹ 1,259,000
Sales required to break even	₹ 4,406,500
Number of unit sales to break even	126
Gross Margin % of Sales	
Gross Profit for the year	₹ 1,800,000
Total Sales for the year	₹ 6,540,000
Contribution Margin	29%
Gross Margin/Total Sales	28%
Total Fixed Expenses	
Operating Expenses	₹ 1,259,000
Breakeven Sales (Annual)	
Gross Margin % of Sales	28%
Total Fixed Expenses	₹ 1,259,000
Yearly Breakeven Amount	₹ 4,574,367
Monthly Breakeven Amount	₹ 381,197

Unit Economics

Financial Statement: https://docs.google.com/spreadsheets/d/1hx_keC0B2tl4alcs2rqSxzl5p275wOrlFDWPtsfTUQ8/edit?usp=sharing

P & L/ unit	Year 2026
Revenue	₹ 36,333
COGS	₹ 25,000
Gross Profit	₹ 11,333
Gross Profit Margin	31%
Operating Costs	₹ 31,994
Operating Profit	₹ 4,338.89

Unit Economics	Year 2026
CAC	₹ 467
CLV	₹ 653
ARPU	₹ 26,160

Step 1	APV = Average Purchase Value = Total revenue/Total number of purchases	₹ 36,333
Step 2	APF = Average Purchase Frequency = Number of purchases/ Number of unique customers	0.72
Step 3	Average Customer Lifespan = 1/Churnrate	0.08
Step 4	CLV =APV*APF*ACL*GM	₹ 653

Assumptions		
Churn rate	Customers at the beginning - Customers at the end/ customers at the beginning	12.5
Customers at the Beginning of month	-	8
Customers at the End of month	-	7

Funding Plan

How much funds required to reach to the next level of the venture?

₹5,00,000

How much have been bootstrapped? If not, why ?

₹3,50,000

How much can be bootstrapped?

₹0 (Further Bootstrapping is not feasible)

How much external funding required? If not, why?

₹5,00,000

Financial Statement: https://docs.google.com/spreadsheets/d/1hx_keC0B2tl4alcs2rqSxzl5p275wOrlFDWPtsfTUQ8/edit?usp=sharing

Funds Utilization Strategy

Category	Description	Revised Annual Amount
R&D and Capital Costs	Product development, equipment, testing tools, and certification compliance.	₹3,00,000
Salaries and HR	Hiring engineers, sales, marketing, and customer service staff to support product development and customer satisfaction.	₹50,000 to ₹1,00,000
Marketing and Promotion	Offline promotions, advertising, and targeted campaigns to build brand awareness and educate the market.	₹20,000 to ₹30,000
Digital Marketing & Customer Service	Building online presence, CRM tools, and dedicated customer support for user queries and technical support.	₹15,000 to ₹25,000
Utilities	Covers essential operational utilities such as electricity and internet.	₹15,000 to ₹20,000
Office Supplies	Consumables and office essentials to support staff and operations.	₹20,000 to ₹30,000
Rent	Office space for R&D, administrative work, and customer support.	₹50,000 to ₹75,000
Contingency Fund	~5-10% of yearly expenses for unforeseen costs (regulatory, equipment, or unexpected operational needs).	~₹30,000 to ₹50,000
Total Fund Utilization	Total funds needed each year to cover operational, marketing, and R&D costs along with contingency allocation.	₹5,00,000

Our Team



SARAVANAN R K

LEAD/CEO

- PROBLEM SOLVING
- IDEATION



VASANTH K

PROJECT MANAGER

- PROJECT MANAGMENT
- DOCUMENTATION



ARAVIND SELVAN C

TECHNICAL ANALYST/CTO

- TECHNICAL ANALYSIS
- PROGRAMMER

Why this Makes Us a Good Team:

- Expertise in tech, clinical care, and regulation.
- Focused on delivering patient-friendly, non-invasive solutions.
- Ready to launch with a strong market and regulatory approach

THANK YOU!

